Chapter 5: Tissues and the Primary Growth of Stems

Did You Know?

- All soft parts of plants (flowers, leaves) are composed of cells that live only briefly (days for flowers, spring to autumn for leaves).
- Most hard parts (wood, seed shells) are composed of cells that usually die immediately after hardening.
- Almost all shoots, even highly modified ones (cacti, orchids, bulbs, vines), have just one pattern of tissues.

Concepts

- The body of an herb consists of three basic parts: **leaves**, **stems**, and **roots**.
- **Evolution of Land Plants**:
  - First land plants evolved about 420 million years ago from algae.
  - Early plants lacked roots, stems, and leaves.
  - Upright growth provided a selective advantage (sunlight) but required transport and support tissues.
  - Stems evolved to transport water/nutrients and support the plant.
- **Functions of Stems**:
  - Transport and support.
  - Produce and hold leaves in sunlight.
  - Store sugars and nutrients (e.g., maples, bulbs, corms).
  - Survival (underground parts remain alive).
  - Dispersal (runners, vines, fragmentation).

Figure 1: (A) The primary body of an herb like this geranium consists of roots, stems, and leaves; buds are located in the axil of each leaf and may grow to be either a vegetative branch or a set of flowers. (B) This Iris is also an herb and never produces wood or bark. Almost all flowering plants are either broadleaf plants (eudicots) like the geranium or plants with grass-like leaves (monocots) like the Iris.

- **Modifications**:
  - Although all flowering plants possess leaves, stems, and roots, these parts have been modified extensively in some species.

- **Cacti**: Stems are photosynthetic and store water; leaves are modified into spines to reduce water loss.

Figure 2: This prickly pear (Opuntia) shows that one plant can have two types of shoot: The "pad" is the main shoot, and the spine clusters are highly modified axillary branch shoots. One of the spine-bearing branches has been stimulated to develop into the first type of shoot and become a pad-like branch. The plant also has two types of leaves: (1) small fleshy green leaves on the young buds and (2) spines on the axillary branches.

- **Orchids**: Some are "shootless" with photosynthetic roots; stems are reduced.
  - **Campylocentrum pachyrrhizum** and **Harrisella porrecta** consist of a mass of green photosynthetic roots connected to a tiny portion of stem.

Figure 3: This orchid carries out photosynthesis by means of roots that contain chlorophyll. The plant grows in trees, attached to their bark (it is epiphytic) such that its roots are exposed to sunlight, they are never in soil. The photosynthetic roots grow in any direction, even upward. The plant has only a small portion of shoot, which produces flowers.

- Most epiphytic orchids resist temporary drying because their shoots are fibrous and have a thick cuticle.

Figure 4: Most epiphytic orchids resist the stresses of temporary drying because their shoots are fibrous and have a thick cuticle composed of cutin and wax.

- **Bromeliads**: Some are nearly rootless, absorbing moisture through leaves from fog (e.g., **Tillandsia straminea**).

Figure 5: These bromeliads are not rooted into the coastal sand dunes—a strong wind can blow them around. All water is absorbed from fog condensing on the leaves.

- **Herbaceous vs. Woody Plants**:
  - **Herb (Primary Plant Body)**: Never becomes woody; covered by epidermis; consists of primary tissues; derived from shoot and root apical meristems.
  - **Woody Plant (Secondary Plant Body)**: Becomes woody; covered by bark; consists of secondary tissues (wood and bark); derived from vascular and cork cambia.

Figure 6: Types of Plant Body: Comparison of Primary plant body (herbaceous) and Secondary plant body (woody).

Basic Types of Cells and Tissues

- All plant cells are grouped into three classes based on their cell walls: **parenchyma**, **collenchyma**, and **sclerenchyma**.

Figure 7: Three Basic Types of Plant Cells and Tissues, Based on Cell Wall: Parenchyma, Collenchyma, and Sclerenchyma.

Parenchyma

- **Characteristics**:
  - Thin primary walls.
  - Typically alive at maturity.
  - Most common type of cell and tissue.

- **Functions**:
  - **Chlorenchyma**: Photosynthetic parenchyma cells with thin walls to allow light and CO_2 to pass.
  - **Glandular cells**: Secrete nectar, fragrances, mucilage, resins, and oils; contain abundant ER and dictyosomes.
  - **Storage**: Store starch, protein, water, etc.
  - **Aerenchyma**: Parenchyma with large intercellular spaces for gas exchange.
  - **Phloem**: Parenchyma tissue that conducts nutrients over long distances.
- **Cost**: Metabolically inexpensive to build due to thin walls.

Figure 8: (A) Parenchyma cells of geranium. Their walls (blue green) are thin, and their vacuoles are large and full of watery contents that did not stain. (B) Chlorenchyma cells from a leaf of privet. Because these cells are small and the section is thicker than that in (A), most cells still have nuclei (red). The structures close to the wall (blue) are chloroplasts. The large white areas are intracellular spaces. (C) Material taken from the center of a pine (Pinus) stem. Cells that have stained dark purple are filled with chemicals called tannins. (D) A resin canal in a pine leaf. The white area is the central cavity where resin is stored, and cells that line the cavity are glandular parenchyma cells.

- **Transfer cells**: Mediate short-distance transport; have wall ingrowths to increase surface area for plasma membrane pumps.

Figure 9: (A) Transfer cells in the salt gland of Frankenia grandifolia. The wall ingrowths increase the surface area of the cell membrane, providing more room for salt-pumping proteins in the membrane. (B) These transfer cells were frozen, then broken open, and all of the protoplasm was washed away with detergent. Only the cell wall is left. All the knobby irregular areas are wall ingrowths.

- **Box 5-1: Plants and People: Parenchyma, Sclerenchyma, and Food**:

  - Most fruits and vegetables are almost pure **parenchyma** (soft, easy to chew, nutritious).
  - **Collenchyma** is found in celery petioles (strings) and rhubarb; used for strength, not usually abundant in food.
  - **Sclerenchyma** (fibers and sclereids) provides texture:
    - Fibers in asparagus, green beans, mangoes.
    - Sclereids cause grittiness in pears and hardness of seed coats (beans, peas, corn).

Collenchyma

- **Characteristics**:
  - Primary walls are unevenly thickened (often at corners).
  - Exhibits **plasticity**: Can be deformed by pressure/tension and retains the new shape.
  - Typically alive at maturity.

Figure 10: (A) Masses of collenchyma cells often occur in the outer parts of stems and leaf stalks. This is part of a Peperomia stem. Collenchyma forms a band about 8 to 12 cells thick. (B) In collenchyma cells, the primary wall is thicker at the corners so the protoplast becomes rounded. No intercellular spaces are present.

- **Location**:
  - Elongating shoot tips (e.g., vines).
  - Layer just under the epidermis or as bands next to vascular bundles.
- **Function**:
  - Provides support to growing tissues.
  - Works with turgid parenchyma (like air pressure in a tire) to support the stem.

Figure 11: Shoot tips of long vines need the plastic support of collenchyma while their stems are elongating.

Sclerenchyma

- **Characteristics**:
  - Primary wall plus a thick, lignified **secondary wall**.
  - Exhibits **elasticity**: Can be deformed but returns to original shape.
  - Many are dead at maturity.
- **Function**:
  - Provides support to mature organs that have stopped growing.
  - Prevents protoplast expansion (cannot grow).
- **Types**:
  - **Mechanical (Nonconducting)**:
    - **Fibers**: Long, flexible; provide elastic support (e.g., wood, bark).
    - **Sclereids**: Short, isodiametric (cuboidal); brittle and inflexible; form hard surfaces (e.g., nut shells, fruit pits).

Figure 12: (A) A mass of fiber cells in the leaf of Agave. These are large, heavy, tough leaves, and fiber masses provide elastic strength. (B) The same mass of fibers as in (A) but viewed with polarized light. Secondary walls shine brightly because their cellulose molecules are packed in a tight, crystalline form. (C) A stem of bamboo was treated with nitric and chromic acid to dissolve the middle lamellas and allow the cells to separate from each other. Fibers are long and narrow; shorter, wider cells are parenchyma. (D) Sclereids are more or less cuboidal, definitely not long like fibers. (E) This portion of a water lily leaf contains large, irregularly branched cells stained red. These are astrosclereids (star-shaped sclereids). (F) A star-shaped sclereid in a water lily leaf shown by scanning electron microscopy.

- **Conducting (Tracheary Elements)**: Transport water (discussed under Xylem).

Figure 13: Types of Sclerenchyma: Mechanical (Sclereids, Fibers) and Conducting (Tracheids, Vessel elements).

- Wood is composed of several types of cells, including fibers for strength and vessel elements for conduction.

Figure 14: Wood is composed of several types of cells. The numerous small cells with thick walls and extremely narrow lumens are fibers, which give wood strength and flexibility. The large round cells that appear to be empty are vessel elements; the small cells with thin walls and large lumens are wood parenchyma cells.

- Some sclerenchyma cells, especially fibers, remain alive at maturity and store starch or calcium oxalate crystals.

Figure 15: These fiber cells have nuclei: They are living cells. Their secondary walls are thick, but not so thick as those shown in Figure 5-11. The small dots in the walls are pits; these are much narrower than the pits shown in Figure 5-10D. Leaf of Smilax.

External Organization of Stems

- **Basic Structure**:
  - **Nodes**: Where leaves are attached.
  - **Internodes**: Regions between nodes.
  - **Leaf axil**: Area just above leaf attachment.
  - **Axillary bud**: Miniature shoot in the leaf axil; can become a branch or flower.
  - **Terminal bud**: At the extreme tip of the stem.
  - **Bud scales**: Modified leaves protecting the bud.

Figure 16: (A) In these buds of pistachio, leaves are modified into bud scales, just as those of prickly pear buds are modified into spines. Bud scales are waterproof and protective, but they are shed when the bud begins to grow in the spring. (B) The two axillary buds of this Viburnum have already begun to grow into branches, and their first young, expanding leaves are visible.

- **Leaf scars**: Where leaves were attached; sealed with cork to prevent water loss and infection.

Figure 17: As a leaf falls from a stem, it breaks cells and creates a wound; a leaf scar is a layer of cork that seals the wound, keeping fungi and bacteria out and preventing water loss.

- **Phyllotaxy** (Leaf Arrangement):
  - **Alternate**: One leaf per node.
  - **Opposite**: Two leaves per node.
  - **Whorled**: Three or more leaves per node.
  - **Distichous**: Leaves in two rows (e.g., corn, iris).
  - **Decussate**: Leaves in four rows (opposite leaves rotated 90 degrees).
  - **Spiral**: Leaves form a spiral up the stem.

Figure 18: Phyllotaxy. (A) These young leaves of redbud (Cercis canadensis) are arranged with alternate phyllotaxy—one leaf per node. (B) Leaves of Salvia show opposite phyllotaxy—two leaves per node, each pair pointing 90 degrees away from previous and subsequent pairs. (C) Irises (Iris) have distichous phyllotaxy—one leaf per node, arranged in just two rows.

Figure 19: Phyllotaxy: Alternate, Opposite, Decussate, Whorled, Spiral, Distichous.

- Internode length varies: Long in vines, short in lettuce/cabbage (leaves packed together).

Figure 20: Cabbage nodes are packed closely together, and all leaves are tightly clustered. Such closely packed, large leaves with spiral phyllotaxy are poor at photosynthesis and did not evolve by natural selection; they were produced by plant geneticists.

- **Stem Modifications**:
  - **Vines**: Long internodes for exploration; may use **tendrils** (modified leaves or branches) for support.

Figure 21: (A) These grape vines (Vitis) place their leaves in the full sun at the top of the forest canopy, even though they have not invested much sugar in building strong trunks. (B) Various vines climb by different methods: This Cocculus grows in a spiral pattern and twines around slender supports. (C) Coral vine (Antigonon leptopus) climbs by means of branches that bend in a zigzag at each node, but then finally produce a tendril that wraps around objects.

- **Stolons (Runners)**: Long, thin internodes; spread along the ground (e.g., strawberry, airplane plant).

Figure 22: Airplane plant (Chlorophytum) is a popular ornamental plant that spreads by runners in nature. Most people are familiar with it as a hanging basket plant, with its runners arching out and then drooping down and forming new plants at the tips. The new plant here has already produced roots.

- **Bulbs**: Short vertical stems with thick, fleshy leaves (e.g., onion).
- **Corms**: Short vertical stems with thin, papery leaves (e.g., gladiolus).
- **Tubers**: Short horizontal stems for storage (e.g., potato).

Figure 23: (A) Onions are bulbs: They have short vertical stems and fleshy leaves. (B) Gladioluses have corms with fleshy stems and papery leaves. (C) A tuber (potato: Solanum tuberosum) is a short horizontal stem that grows only for a limited period of time. Its leaves are microscopic, and its axillary buds are the potato's "eyes."

- **Rhizomes**: Fleshy horizontal underground stems (e.g., iris, bamboo).
- **Division of Labor**: Plants often have multiple shoot types (e.g., rhizomes for spread/ survival, aerial shoots for photosynthesis).

Figure 24: (A) Division of labor. False Solomon's seal (Smilacena racemosa) consists of subterranean rhizomes (which survive harsh winters and spread through the soil) and aerial shoots (which carry out photosynthesis and flowering). (B) This succulent spurge (Euphorbia) has two types of shoots: the long wide green shoot that lives for many years, and many narrow, red shoots that die quickly then function as spines.

- Axillary buds do not activate at random; patterns exist (e.g., dormancy in shade to favor vertical growth).

Figure 25: (A) In a heavily shaded forest, many axillary buds remain permanently dormant, allowing all resources to be concentrated on placing one shoot up into the light. (B) With abundant light, one main trunk is not so advantageous; instead, many axillary buds grow as branches, allowing rapid production of leaves.

- **Box 5-2: Organs: Replace Them or Reuse Them?**:
  - Animals typically have one set of organs (heart, brain) for life.
  - Plants have many of each organ (leaves, roots) and replace them periodically (“one-use” organs).
  - Advantages: Replaces damaged/dirty organs; allows for new, healthy tissues (e.g., new vascular system every year in wood).

Internal Organization of Stems: Arrangement of Primary Tissues

Epidermis

- **Structure**: Single layer of living parenchyma cells.
- **Function**:
  - Barrier against water loss, pathogens, and herbivores.
  - Gas exchange.
- **Modifications**:
  - **Cutin/Cuticle**: Fatty substance making the wall impermeable to water.

Figure 26: (A) The cuticle on this ivy (Hedera helix) has been stained dark pink; the lighter layer is wall material impregnated with cutin. Epidermal protoplasts have stained dark blue. Interior cells are photosynthetic collenchyma. (B) This epidermis was taken from a desert plant, Agave. Its cuticle is much thicker than that of ivy in (A), and the outer walls of the epidermal cells are extremely thick.

- **Wax**: Additional waterproofing layer; indigestible to insects.

Figure 27: On this epidermis, wax is present as flat plates, but in other species it can occur as threads, beads, flakes, or liquid.

Figure 28: Special Chemicals in Walls: Collenchyma (Pectins), Sclerenchyma (Lignin), Epidermis (Cutin/waxes), Endodermis (Suberin, Lignin), Cork (Suberin).

- **Stomata**: Pores guarded by **guard cells** for gas exchange (CO_2 entry, H_2O exit).

Figure 29: (A) Light microscope view of the epidermis of Zebrina; the stomatal pore, guard cells, and adjacent cells are visible, as are ordinary epidermal cells. (B) Scanning electron micrograph of the epidermis of a cactus showing numerous stomata. (C) The stomatal pore is surrounded by guard cells and then other epidermal cells. Notice the small particle of debris at one end of the pore.

- Guard cells swell (turgid) to open the pore and shrink (flaccid) to close it.

Figure 30: (A) When guard cells absorb water and swell, they become curved and the inner walls are pulled apart by the expansion of the back walls. (B) If guard cells lose water and shrink, they straighten, closing the stomatal pore.

- **Trichomes (Hairs)**:
  - Protection from herbivores (physical or chemical).
  - Shade the leaf.
  - Reduce water loss by creating a boundary layer of air.

Figure 31: Trichomes. (A) Flat, scale-shaped trichome. (B and C) Branched trichomes. (D) Glandular trichomes—these are two large single cells. (E) Glandular trichome with a stalk cell elevating the secretory head cell. (F) Glandular trichome, capable of injecting poison. (G and H) Simple, unbranched, nonglandular trichomes.

Cortex

- **Location**: Interior to the epidermis.
- **Composition**: Mostly parenchyma (photosynthetic, storage), sometimes collenchyma.
- **Specialized cells**: Secretory cells (latex, resin), crystal-containing cells.
- **Aerenchyma**: Cortex with large air spaces (aquatic plants).

Figure 32: (A) The cortex of this stem of buttercup (Ranunculus) is the narrow band of cells between the epidermis and the vascular bundles. (B) The cortex of this corn stem is even narrower, in some places only two or three cells wide.

- **Box 5-3: Alternatives: Simple Plants**:
  - **Hornworts** and **Liverworts** have simple bodies (thin sheets of parenchyma).
  - Lack epidermis, stomata, xylem, phloem.
  - Absorb water/minerals directly into cells.
  - **Mosses** are slightly more complex, with stems and leaves; some have conducting cells (**hydroids** and **leptoids**).

Figure 33: Plants of hornworts (Phaeoceros); the bodies are just thin, green sheets of parenchyma. The columns ("horns") are reproductive structures.

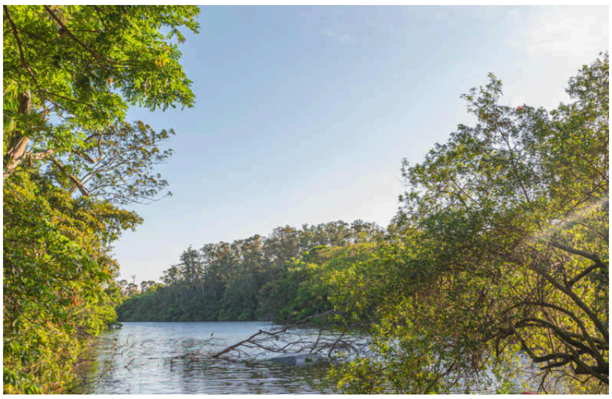

Figure 34: Plants of this liverwort (Riccia fluitans) grow in quiet streams; each shoot is less than 1 mm thick and consists of parenchyma cells. Each grows from an apical meristem that occasionally divides in two.

Vascular Tissues

- **Xylem**: Conducts water and minerals.
- **Phloem**: Distributes sugars and minerals.

Xylem

- **Conducting Cells (Tracheary Elements)**: Dead at maturity; hollow secondary walls.
  - **Tracheids**:
    - Long, narrow, tapered ends.
    - No perforations; water moves through **pit-pairs**.
    - Found in all vascular plants.
  - **Vessel Elements**:
    - Short, wide, flat ends.
    - Have **perforations** (holes) in end walls.
    - Stack to form **vessels**.
    - Found almost exclusively in angiosperms.

Figure 35: Tracheary elements. (A, C, and D) Tracheids are long cells with tapered ends. The primary wall is complete over the whole surface, and in (C), it is so thin that the inner, spiral secondary wall is visible. (B and E) Vessel elements tend to be wider and shorter than tracheids, but the most important feature is the perforation, the large hole at each end. (F) The perforation of one vessel element must be aligned with that of the next if water is to pass through with little friction; the stack of vessel elements is a vessel.

- **Secondary Wall Patterns**:
  - **Annular**: Rings; weak but extensible (Protoxylem).
  - **Helical**: Spiral; extensible (Protoxylem).
  - **Scalariform**: Ladder-like; stronger.
  - **Reticulate**: Net-like.
  - **Circular Bordered Pits**: Strongest; rigid (Metaxylem).

Figure 36: Tracheids. (A–D) The growth of a young cytoplasmic cell into a tracheid. After the cell has almost reached mature size, the secondary wall is deposited interior to the primary wall (C). (D) A tracheid cut open, showing a secondary wall in the form of helical thickenings. (E) A whole cell in which the primary wall is so thin that the annular secondary walls inside are visible. (F) Whole cell with helical secondary wall. (G) Scalariform secondary wall. (H) Reticulate secondary wall. (I) Pitted secondary wall. (J) Light micrograph of a tracheid with a helical secondary wall.

Figure 37: Tracheary Elements (Conducting Sclerenchyma): Comparison of Tracheids and Vessel Elements.

- Water movement causes inward traction on walls; secondary walls prevent collapse.

Figure 38: (A) As water enters a tracheary element, it passes through primary wall between regions of secondary wall. (B) Water adheres to cellulose molecules, pulling walls inward and upward as it rises in xylem; primary walls tend to collapse but secondary walls provide strength to hold the primary wall in place. (C) Under severe water stress (dry air and dry soil), annular and helical secondary walls may not be strong enough to prevent collapse of the primary wall.

- **Pits**: Areas where secondary wall is absent.
  - **Bordered pits**: Have a rim of extra wall material for strength.

Figure 39: (A) Side view of a pit that has been bisected. In a sclereid or fiber, the secondary wall would be thicker and the pit narrower and deeper. (B) A bordered pit has a rim of extra wall material that prevents the pit from being a weak spot in the wall. With borders, pits can be broad enough to allow rapid flow of water from tracheary element to tracheary element, but not so weak as to be detrimental.

Figure 40: (A) Bordered pits in face view on front wall of a vessel element. The white slits are the actual openings of the pits, the pinkish area around each white region is the border, and the dark red hexagons are regions where the secondary wall lies against the primary wall. (B) Bordered pits in face view on the front walls of tracheids of pine. (C) Scalariform bordered pits.

- **Perforations**: Complete holes in primary and secondary walls of vessel elements; reduce friction.

Figure 41: This vessel element was isolated from tissue by dissolving the middle lamella. The secondary wall (the network along the cell's end and sides) is clearly visible on the end because the primary wall was digested away before the cell died; these open holes constitute a perforation. The thin, almost transparent covering on the cell's sides is the primary wall, which was not digested.

- Tracheids obtain water from others through pit-pairs; vessels through perforations (end-to-end) and pits (lateral).

Figure 42: (A) Water passes between tracheids only through pit-pairs. Other than plasmodesmata, no holes are present in primary walls of tracheids. (B) Water can pass between vessel elements through perforations, but it can pass from one vessel to another only through pit-pairs. Vessels are long, but not as long as a whole plant, and water must, therefore, pass from one vessel to another as it moves upward from roots to stems to leaves.

- Ferns have tracheids, but some have evolved crude perforations.

Figure 43: This fern (Astrolepis) tracheary element had digested portions of primary wall out of several pits, converting them to crude perforations. In angiosperms, the wall-digesting mechanism is more refined and removes the primary wall more completely.

Figure 44: Length of Vessels in Trunks of Trees: Average and Maximum lengths for various species.

Phloem

- **Conducting Cells (Sieve Elements)**: Alive at maturity; primary walls only.
  - **Sieve Cells**:
    - Long, narrow, tapered ends.
    - Small **sieve pores** over surface.
    - Associated with **albuminous cells**.
    - Found in non-angiosperm vascular plants.
  - **Sieve Tube Members**:
    - Short, wide, flat ends.
    - **Sieve plates** with large pores on end walls.
    - Stack to form **sieve tubes**.
    - Associated with **companion cells** (load/unload sugars, control metabolism).
    - Found in angiosperms.

Figure 45: (A) A sieve cell is shaped like a tracheid and has sieve areas with sieve pores over much of its surface. (B) A sieve tube member has flat end walls (sieve plates) with large sieve pores. On the side walls are only a few small sieve areas with very narrow sieve pores.


Figure 46: Large sieve areas present on sieve plates of sieve tube members. Phloem sap flows from one tube-like sieve tube member into the next through the wide sieve pores.


Figure 47: Sieve Elements: Comparison of Sieve Cells and Sieve Tube Members.

- **Companion Cells**: Small, rich in cytoplasm/ribosomes; control metabolism of sieve tube members.

Figure 48: (A) In a vascular bundle, sieve tube members and companion cells occur together. Companion cells are small and rich in cytoplasm; sieve tube members are large and appear empty because most phloem sap is lost while preparing the tissue for microscopy. (B) Electron micrograph showing two companion cells and two sieve tube members. (C) The connection between a companion cell and a sieve tube member is a primary pit field with plasmodesmata on the side of the companion cell, but it is a sieve area with sieve pores on the side of the sieve tube members.

Vascular Bundles

- **Composition**: Xylem and phloem strands running parallel.
- **Arrangement**:
  - **Eudicots**: Ring of bundles surrounding pith.
  - **Monocots**: Complex network of scattered bundles.
- **Collateral**: Xylem and phloem side-by-side.
- **Primary Xylem**: Contains protoxylem (inner) and metaxylem (outer).
- **Primary Phloem**: Contains protophloem (outer) and metaphloem (inner).

Figure 49: (A) In this vascular bundle of sunflower (Helianthus), xylem contains rows of vessels alternating with rows of xylem parenchyma. Phloem contains sieve tube members and companion cells. There is a large sclerenchyma cap of phloem fibers. (B) Vascular bundles of monocots have a more complex arrangement. In this corn stem, many large bundles are located near the cortex, and fewer, smaller bundles in the ground tissue. (C) High magnification of a corn bundle, showing primary xylem and phloem.

Stem Growth and Differentiation

- **Apical Meristem**: Region of cell division at the shoot tip.
- **Subapical Meristem**: Region below apical meristem where differentiation begins.

Figure 50: (A) This shoot tip has several fully expanded leaves, one set of small leaves, and another set of even smaller, younger leaves. (B) Longitudinal section through a shoot tip of Coleus showing the apical meristem and leaf primordia of various ages. (C) Higher magnification of the Coleus shoot apical meristem. Vascular tissue is differentiating in the center of each leaf primordium. (D) This apical meristem of Elodea is much taller and narrower than that of Coleus in (C), and it has many more leaf primordia. (E) This is the giant shoot apical meristem of golden barrel cactus (Echinocactus grusonii).

- **Differentiation Sequence**:
  - **Protoxylem/Protophloem**: Differentiate while stem is elongating; must be extensible (annular/helical walls).
  - **Metaxylem/Metaphloem**: Differentiate after elongation stops; can have rigid walls (pitted).

Figure 51: Development of three tracheary elements. (A) All cells are young, small, and cytoplasmic. (B) The first protoxylem element forms while all cells are still small. (C) Cells 2 and 3 continue growing, stretching cell 1, which is dead. (D and E) Cell 2 matures while cell 3 continues to grow, stretching cells 1 and 2. (F) All elongation has stopped, and cell 3 differentiates as a pitted element. The primary wall of cell 1 has been torn and no longer conducts. Cell 2 is probably still capable of carrying water.

- **Meristematic Tissues**:
  - **Protoderm** -> Epidermis.
  - **Provascular tissue** -> Primary Xylem and Phloem.
  - **Ground meristem** -> Pith and Cortex.

Figure 52: Cells of an apical meristem give rise to three types of subapical cells, which in turn give rise to particular types of mature primary tissues. Epidermis develops only from protoderm, never from provascular tissue or ground meristem; other cell lineages in the primary body are also quite simple and linear.

- **Box 5-4: Plants and People Grow Differently**:
  - Animals have **determinate growth** (fixed size/number of organs).
  - Plants have **indeterminate growth** (grow indefinitely, no set number of organs).
  - **Determinate organogenesis** (animals) vs. **Indeterminate organogenesis** (plants).
  - Exceptions: Annual plants have determinate growth; Tapeworms have indeterminate growth.